

Lezione del 10-4-2026

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Osserviamo che se

$$Z = |Z|(\cos \theta + i \sin \theta)$$

$$W = |W|(\cos \tau + i \sin \tau)$$

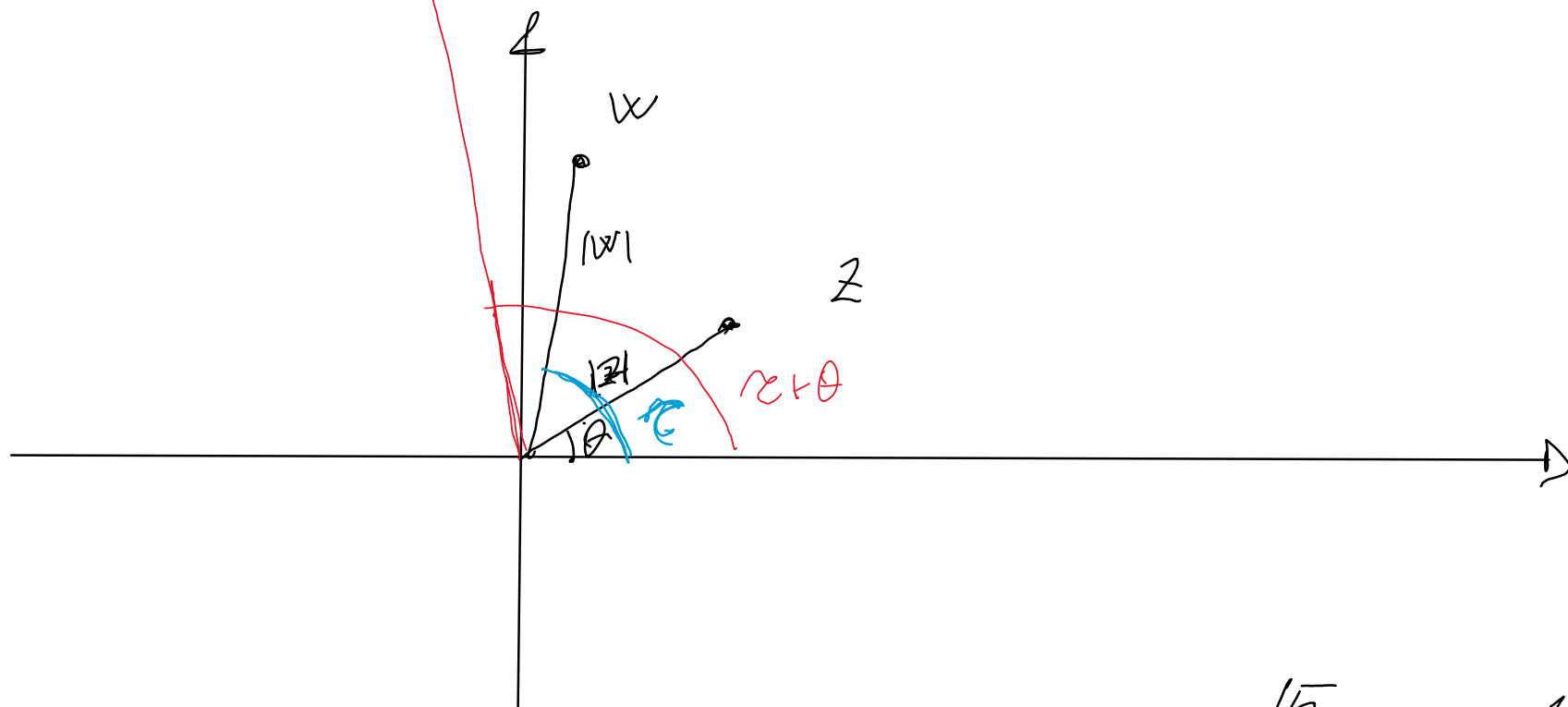
e il calcolo è

$$Z \cdot W = |Z|(\cos \theta + i \sin \theta) \cdot |W|(\cos \tau + i \sin \tau)$$

$$= |Z||W|(\cos \theta \cos \tau + i \cos \theta \sin \tau + i \sin \theta \cos \tau + i^2 \sin \theta \sin \tau)$$

$$= |Z||W|(\cos \theta \cos \tau - \sin \theta \sin \tau + i(\cos \theta \sin \tau + \sin \theta \cos \tau))$$

$$= |z|/|w| \left(\cos(\theta + \varphi) + i \sin(\theta + \varphi) \right)$$



In particolare, z^n $z = 1 + i$ $z^{15} = (1+i)^{15}$

$$z = |z| (\cos \theta + i \sin \theta)$$

$$z^n = (|z|)^n (\cos n\theta + i \sin n\theta)$$

∞

$$(|z|(\cos \theta + i \sin \theta)) = |z| (\cos(n\theta) + i \sin(n\theta))$$

Ex

$$(1+i)^{15}$$

$$z = 1+i \rightarrow |z| = \sqrt{2}$$

$$\theta: \begin{cases} 1 = \sqrt{2} \cos \theta \\ 1 = \sqrt{2} \sin \theta \end{cases} \rightarrow \begin{cases} \cos \theta = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \\ \sin \theta = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \end{cases}$$

$$\theta = \frac{\pi}{4} + 2k\pi$$

$$z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$z^{15} = (\sqrt{2})^{15} \left(\cos \left(15 \frac{\pi}{4} \right) + i \sin \left(15 \frac{\pi}{4} \right) \right)$$

$$= 2^{\frac{15}{2}} \left(\cos \left(\frac{7}{4}\pi + 2\pi \right) + i \sin \left(\frac{7}{4}\pi + 2\pi \right) \right) =$$

$$= 2^7 \sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \left(\frac{7\pi}{4} \right) \right) =$$

$$= 2^7 \sqrt{2} \left(\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) = 2^7 (1 - i)$$

$$15 \frac{\pi}{4} = (12 + 3) \frac{\pi}{4} = 3\pi + \frac{3\pi}{4} = 2\pi + \pi + \frac{3\pi}{4}$$

$$= \frac{7\pi}{4} + 2\pi$$

oss

$$\frac{7\pi}{4} = 2\pi - \frac{\pi}{4}$$

$$\cos \left(\frac{7\pi}{4} \right) = \cos \left(2\pi - \frac{\pi}{4} \right) = + \cos \left(\frac{\pi}{4} \right) = \frac{\sqrt{2}}{2}$$

$$\sin \left(\frac{7\pi}{4} \right) = \sin \left(2\pi - \frac{\pi}{4} \right) = - \sin \left(\frac{\pi}{4} \right) = - \frac{\sqrt{2}}{2}$$

✓

✓

Def Due di $\frac{z}{w}$! $w \neq 0$

(usare le forme trigonometriche)

Quando due numeri complessi z e w ,
scritti in forme trigonometriche, sono UGUALI

$$z = |z| (\cos \theta + i \sin \theta)$$

$$w = |w| (\cos \varphi + i \sin \varphi)$$

$$z = w ? \Leftrightarrow \begin{cases} |z| = |w| \\ \theta = \varphi + k \cdot 2\pi \end{cases} \Leftrightarrow \begin{cases} |z| = |w| \\ \theta - \varphi = k \cdot 2\pi \end{cases}$$

$$|z| = r + n \omega \quad \omega \quad \omega - n \quad \omega$$

$$\sqrt{3} \left(\cos \frac{13\pi}{5} + i \sin \frac{13\pi}{5} \right) = z \quad \Rightarrow z = \omega$$

$$\sqrt{3} \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right) = \omega$$

$$\frac{13\pi}{5} - \frac{3\pi}{5} = \frac{10\pi}{5} = 2\pi$$

EQUAZIONE PONDERIA IN $\mathbb{C}$

$\sum \mathbb{R}$

$$x^n = 0$$

Levi azione su eq. monomiale.

$$\text{In } \mathbb{C} \quad z^n = w \quad \begin{matrix} n \in \mathbb{N} \\ w \in \mathbb{C} \end{matrix}$$

T (insieme delle radici n -me complesse)

Se $w \in \mathbb{C}$, allora esistono

n differenti radici n -me dette

da:

$$\sqrt[n]{r} \cdot \left(\cos \frac{\theta + k2\pi}{n} + i \sin \frac{\theta + k2\pi}{n} \right)$$

$$z_k = \sqrt[n]{|w|} \left(\cos \frac{\theta}{n} + i \sin \frac{\theta}{n} \right)$$

$$k=0, 1, 2, \dots, n-1$$

$$(z^n = w \text{ significa } z^n = w)$$

DIM

Se $w \in \mathbb{C}$, scriverlo in forma trigonometrica.

$$w = |w| (\cos \theta + i \sin \theta)$$

Se z è una delle radici n^{me} complesse
 di w , allora le altre radici sono $z \cdot \omega^k$ per $k=1, 2, \dots, n-1$

allora $z = w$! (per definizione)

h. scriviamo $z = |z| (\cos z + i \sin z)$

$\Rightarrow$

$$z^n = |z|^n (\cos(nz) + i \sin(nz))$$

Per tanto $z^n = w \Leftrightarrow$

$$\begin{cases} |w| = |z^n| = |z|^n \\ \theta = nz + k2\pi \end{cases}$$

$$\Rightarrow \begin{cases} |z| = \sqrt[n]{|w|} \quad \text{in } \mathbb{R} \\ z = \frac{\theta + 2\pi k}{n} \\ k \in \mathbb{Z} \end{cases}$$

Per tanto

Unimodular

$$Z = \sqrt[n]{|w|} \left(\cos \frac{\theta + 2\pi k}{n} + i \sin \frac{\theta + 2\pi k}{n} \right)$$

Si prova che sono DISTINTI i
valori di seno e coseno degli angoli

$$\frac{\theta + k 2\pi}{n}, \quad k = 0, 1, 2, \dots, n-1$$

$$\frac{\theta + 2\pi n}{n} = \frac{\theta}{n} + 2\pi$$

$$\frac{\theta + 2\pi}{n} = \frac{\theta + (n+1)2\pi}{n} = \frac{\theta + 2\pi + 2n\pi}{n}$$

$$z_k = \sqrt[n]{|w|} \left(\cos \frac{\theta + k2\pi}{n} + i \sin \frac{\theta + k2\pi}{n} \right)$$

$k=0, 1, 2, \dots$

E+

Calcolare le radici terze di

$$1 + i$$

(risolvere l'eq. $z^3 = 1+i$)

$$w = 1+i$$

$$1+i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

le $\sqrt[3]{1+i}$ sono

$$z_k = \sqrt[3]{\sqrt{2}} \left(\cos \left(\frac{\frac{\pi}{4} + k2\pi}{3} \right) + i \sin \left(\frac{\frac{\pi}{4} + k2\pi}{3} \right) \right)$$

$$k = 0, 1, 2$$

$$z_0 = \sqrt[3]{\sqrt{2}} \left(\cos \left(\frac{\frac{\pi}{4} + 0}{3} \right) + i \sin \left(\frac{\pi}{12} \right) \right)$$

$$2 \sqrt[3]{\sqrt{2}} \left(\cos \left(\frac{\pi}{4} + 2\pi \right) + i \sin \left(\frac{\pi}{4} + 2\pi \right) \right) \rightarrow$$

$$z_1 = \sqrt[6]{2} \left(\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right)$$

$$= \sqrt[6]{2} \left(\cos\left(\frac{9}{12}\pi\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$$

$$z_2 = \sqrt[6]{2} \left(\cos\left(\frac{\frac{\pi}{4} + 4\pi}{3}\right) + i \sin\left(\frac{\frac{\pi}{4} + 4\pi}{3}\right) \right)$$

$$= \sqrt[6]{2} \left(\cos\frac{17}{12}\pi + i \sin\left(\pi + \frac{5}{12}\pi\right) \right)$$

$\cos\frac{\pi}{12} \rightarrow$ dalle formule di BISEZIONE

$$\cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}$$

$$\left(\sin^2 \alpha = \frac{1 - \cos 2\alpha}{2} \right)$$

$$\cos^2\left(\frac{\pi}{12}\right) = \frac{1 + \cos \frac{\pi}{6}}{2} = \frac{1 + \frac{\sqrt{3}}{2}}{2} = \frac{2 + \sqrt{3}}{4}$$

$$\Rightarrow \cos\left(\frac{\pi}{12}\right) = \sqrt{\frac{2 + \sqrt{3}}{4}}$$

↑
I Quadrante

$$\sin\left(\frac{\pi}{12}\right) = \sqrt{\frac{2 - \sqrt{3}}{4}}$$

$$\cos\left(\frac{17}{12}\pi\right) - \cos\left(\pi + \frac{\pi}{12}\right) = -\cos\left(\frac{5}{12}\pi\right)$$

$$\cos\left(\frac{5\pi}{12}\right) = \cos\left(\frac{\pi}{3} + \frac{\pi}{12}\right)$$

$$\cos\left(\frac{\pi}{12}\right)$$

$$\cos\left(\frac{5\pi}{12}\right) = \cos\left(\frac{\pi}{3} + \frac{\pi}{12}\right) = \cos\left(\frac{\pi}{3}\right)\cos\left(\frac{\pi}{12}\right) -$$

$$\sin\left(\frac{\pi}{3}\right)\sin\left(\frac{\pi}{12}\right)$$

$$\frac{5}{12} = \frac{4+1}{12} = \frac{1}{3} + \frac{1}{12}$$

Vale il seguente risultato:

Le due RADICE n^{mo} dell'UNITÀ

(le n^{a} e 1), sono le soluzioni

$$x^n = 1$$

una risposta $z - I$

Se z_i è una QUALSIASI RADICE n^{me}
 di W , allora

TUTTE LE RADICI n^{me} di W sono
 dato da

$$z_k = z_i \cdot \zeta_k, \quad k = 0, 1, \dots, n-1$$

dove ζ_k sono le RADICI n^{me} dell'UNITÀ

Ritornando all'esempio, z_1 è un' unica individuazione

$$z_1 = \sqrt[6]{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) =$$

$$= \sqrt[6]{2} \left(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right)$$

Calcolo le radici TERZE dell'UNITÀ

$$1 = |1| (\cos 0 + i \sin 0) \neq$$

$$\rho_k = |1| \left(\cos \frac{0 + k2\pi}{3} + i \sin \frac{0 + k2\pi}{3} \right)$$

$-k$

$$k = 0, 1, 2$$

$$\rho_0 = 1$$

$$\rho_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\rho_2 = \cos \left(\frac{4\pi}{3} \right) + i \sin \left(\frac{4\pi}{3} \right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

e quindi

$$Z = \sqrt[6]{2} \left(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) \cdot 1$$

$$z_1 = \sqrt[6]{2} \left(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) \left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i \right)$$

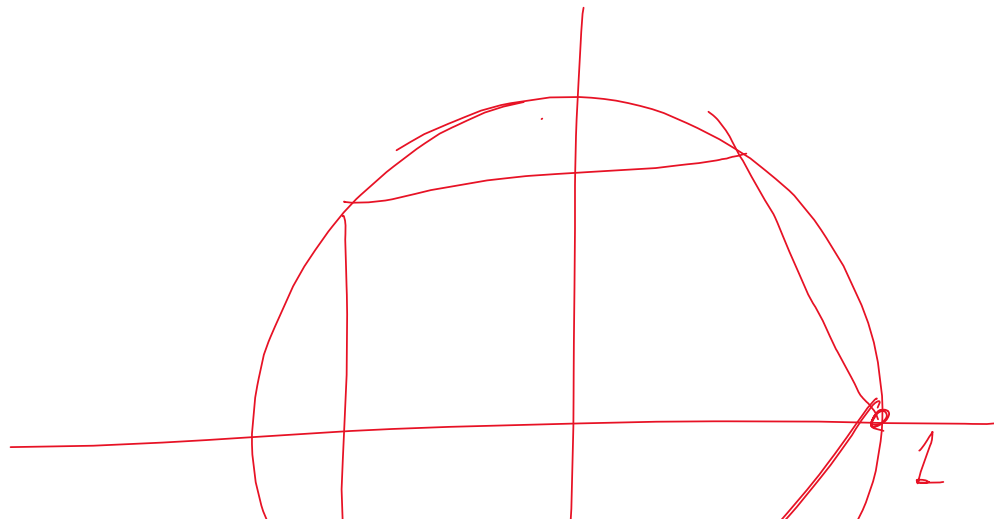
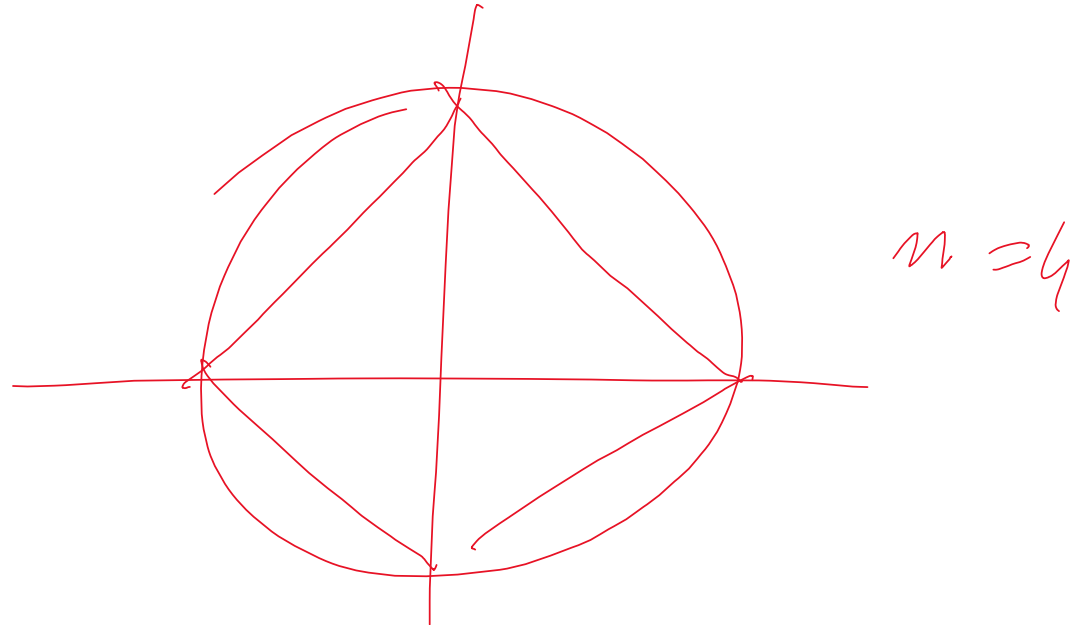
$$z_2 = \sqrt[6]{2} \left(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right) \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right)$$

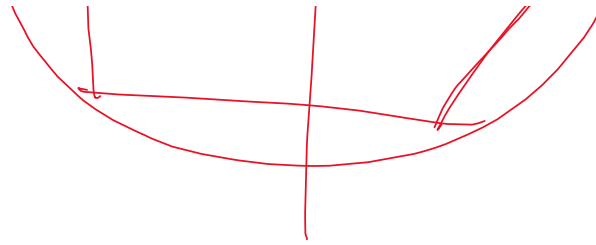
n	RADICI UNITA'
2	± 1
3	$1 ; -\frac{1}{2} + \frac{\sqrt{3}}{2} i ; -\frac{1}{2} - \frac{\sqrt{3}}{2} i$
4	$\pm 1 ; \pm i$



OSS Condizionando le n RADICI n^{me}
dell'unità nel piano (di Argand-Goursat)
esse risultano i VERTICI di un
poligono REGOLARE di n lati
(INSCRITTO nella circonferenza
di RAGGIO 1)







l'equazione di 2° grado

$$a z^2 + b z + c = 0 \quad a, b, c \in \mathbb{C}$$

ha per soluzioni

$$z_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a} \quad (\Delta \in \mathbb{C})$$

Ex

$$z^2 + (1-i)z + 2-2i = 0$$

$$Z_{1,2} = \frac{-(1-i) \pm \sqrt{(1-i)^2 - 4(1)(2-2i)}}{2(1)}$$

$$= \frac{-(1-i) \pm \sqrt{1-1-2i-8+8i}}{2} =$$

$$= \frac{-(1-i) \pm \sqrt{-8+6i}}{2}$$

$$\Delta = -8 + 6i$$

$$|-8+6i| = \sqrt{64+36} = \sqrt{100} = 10$$

$$\Theta: \begin{cases} -8 = 10 \cos \Theta \\ 6 = 10 \sin \Theta \end{cases} \rightarrow \begin{cases} \cos \Theta = -\frac{4}{5} \\ \sin \Theta = \frac{3}{5} \end{cases} \quad \begin{array}{l} \text{NO} \\ \text{ARCHI} \\ \text{NOTI} \end{array}$$

$$\sqrt{-8+6i} = \sqrt{10} \left(\cos \bar{\Theta} + i \sin \bar{\Theta} \right)$$

$$\sqrt[n]{\Delta_{1,2}} = \sqrt[n]{10} \left(\cos \left(\frac{\bar{\Theta} + k2\pi}{n} \right) + i \sin \left(\frac{\bar{\Theta} + k2\pi}{n} \right) \right)$$

$$\sqrt[n]{\Delta_1} = \sqrt[n]{10} \left(\cos \left(\frac{\bar{\Theta}}{n} \right) + i \sin \left(\frac{\bar{\Theta}}{n} \right) \right) \quad k=0,1$$

$$\left(\sqrt{\Delta_2} = \sqrt{10} \left(\cos \frac{\bar{\theta} + 2\pi}{2} + i \sin \left(\frac{\bar{\theta} + 2\pi}{2} \right) \right) \right)$$

$$\cos \left(\frac{\bar{\theta}}{2} \right) =$$

$$\cos \bar{\theta} = -\frac{4}{5}$$

$$\sin \bar{\theta} = \frac{3}{5}$$

$$\cos \left(\frac{\theta}{2} \right) = \frac{1}{2} \cos \theta$$

$$\cos^2 \left(\frac{\bar{\theta}}{2} \right) = \frac{1 + \cos \bar{\theta}}{2} = \frac{1 + \left(-\frac{4}{5} \right)}{2}$$

$$\sin^2 \left(\frac{\bar{\theta}}{2} \right) = \frac{1 - \cos \bar{\theta}}{2} = \frac{1 - \left(-\frac{4}{5} \right)}{2}$$

$$\cos(\frac{\theta}{2}) = \frac{1}{2} = \frac{1}{2}$$

$$\cos^2(\frac{\theta}{2}) = \frac{1}{10}$$

$$\sin^2(\frac{\theta}{2}) = \frac{2}{10} = \frac{1}{5}$$

$$\cos(\frac{\theta}{2}) = \pm \sqrt{1/10}$$

$$\sin(\frac{\theta}{2}) = \pm \sqrt{1/5}$$

$$\sqrt{\Delta_1} = \sqrt{10} \left(\frac{1}{\sqrt{10}} + i \frac{1}{\sqrt{5}} \right) = 1 + \sqrt{2} i$$

$$\sqrt{\Delta_2} = -(1 + \sqrt{2} i)$$

$$\frac{\sqrt{10}}{\sqrt{5}} = \frac{\sqrt{2+5}}{\sqrt{5}} = \frac{\sqrt{2}}{1}$$

$$z_{1,2} = \frac{-(1-i) \pm (1+\sqrt{2}i)}{2} = \dots$$

OSS

$$\begin{cases} \cos \theta = \frac{4}{5} \\ \sin \theta = -\frac{3}{5} \end{cases} \rightarrow \frac{\frac{3}{2}\pi}{2} \leq \frac{\theta}{2} \leq \frac{2\pi}{2}$$

$$\frac{3}{4}\pi \leq \frac{\theta}{2} \leq \pi$$

$$\cos^2\left(\frac{\theta}{2}\right) = \frac{1 + \cos \theta}{2} \rightarrow \cos^2\left(\frac{\theta}{2}\right) = \frac{1 + \frac{4}{5}}{2}$$

$$\cos \frac{\theta}{2} = \frac{1 + \cos \theta}{2} \quad \cos \frac{\theta}{2} = \frac{1 + \frac{4}{5}}{2}$$

$$\cos\left(\frac{\theta}{2}\right) = \frac{1 - \cos\theta}{2}$$

$$\cos\left(\frac{\theta}{2}\right) = \frac{1}{2}$$

$$\cos\left(\frac{\theta}{2}\right) = \frac{8}{10} \rightarrow \cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{9}{10}}$$

$$\sin\left(\frac{\theta}{2}\right) = \frac{8}{10}$$

$$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{4}{5}}$$